

课堂笔记整理 4.10

例 7-8

$$\begin{aligned} \text{可得 } g(t) &= 3u(-t) - 2[u(t) - u(t-1)] + (2t-4)u(t-1) \\ &= 3u(-t) - 2u(t) + (2t-4+2)u(t-1) \\ &= 3u(-t) - 2u(t) + 2r(t-1) \\ &= 3[1-u(t)] - 2u(t) + 2r(t-1) = 3 - 5u(t) + 2r(t-1) \end{aligned}$$

练 7-8

$$\begin{aligned} h(t) &= -4[u(t) - u(t-2)] + (3t-8)[u(t-2) - u(t-6)] \\ &= -4u(t) + 4u(t-2) + (3t-2)u(t-2) - (3t-6+10)u(t-6) \\ &= -4u(t) + 4u(t-2) + 3r(t-2) - 2u(t-2) - 3r(t-6) - 10u(t-6) \\ &= -4u(t) + 2u(t-2) + 3r(t-2) - 3r(t-6) - 10u(t-6) \end{aligned}$$

例 7-10

$$\begin{aligned} \text{当 } t < 0 \text{ 时, } v(0^-) &= \frac{5}{5+3} \times 24 = 15 \text{ V}, v(0) = v(0^-) = v(0^+) = 15 \text{ V} \\ \text{当 } t > 0 \text{ 时, } R_{th} &= 4 \text{ k}\Omega, \tau = R_{th}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s} \\ v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} = 30 + (15-30)e^{-\frac{t}{2}} = 30 - 15e^{-0.5t} \text{ V} \\ \text{当 } t=1 \text{ 时, } v(1) &= 30 - 15e^{-0.5} = 20.902 \text{ V} \\ \text{当 } t=4 \text{ 时, } v(4) &= 30 - 15e^{-2} = 27.97 \text{ V} \end{aligned}$$

练 7-10

$$\begin{aligned} v(0) &= 15 \text{ V}, v(\infty) = 15 - ((15+2.5)/8) \times 2 = 9.375 \text{ V} \\ R &= 2116 = 1.5 \Omega, \tau = RC = \frac{3}{2} \times \frac{1}{5} = \frac{1}{2} \\ v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} \\ &= (9.375 + [15 - 9.375]e^{-\frac{t}{1/2}}) \text{ V} \\ &= (9.375 + 5.625e^{-2t}) \text{ V} \\ v(0.5) &= 9.375 + \frac{5.625}{e} = 11.444 \text{ V} \end{aligned}$$

例 7-11

$$\begin{aligned} \text{当 } t < 0 \text{ 时, } v &= 10 \text{ V}, i = -\frac{v}{10} = -1 \text{ A}, v(0) = v(0^-) = 10 \text{ V} \\ \text{当 } t > 0 \text{ 时, } v(\infty) &= \frac{20}{20+10} \times 30 = 20 \text{ V}, R_{th} = 10 \parallel 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega \\ \tau &= R_{th}C = \frac{20}{3} \times \frac{1}{4} = \frac{5}{3} \text{ s} \\ \text{因此 } v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} = 20 + (10-20)e^{-\frac{3}{5}t} = 20 - 10e^{-0.6t} \text{ V} \\ i &= i_{un} + i_c = \frac{v}{20} + C \frac{dv}{dt} = 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} \\ &= (1 + e^{-0.6t}) \text{ A} \\ \text{所以 } v &= \begin{cases} 10 \text{ V} & t < 0 \\ (20 - 10e^{-0.6t}) \text{ V} & t \geq 0 \end{cases} \quad i = \begin{cases} -1 \text{ A} & t < 0 \\ (1 + e^{-0.6t}) \text{ A} & t \geq 0 \end{cases} \end{aligned}$$

练7-11

$$V_0 = 20V, I_0 = 0, t > 0, v(\infty) = \frac{V}{15} \times 30 = 10V, R_{eq} = \sqrt{110} = \frac{10}{3}\Omega$$

$$\tau = 0.2 \times \frac{10}{3} = \frac{2}{3}s$$

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}} = (10 + [20 - 10]e^{-\frac{3}{2}t})V = 10(1 + e^{-1.5t})V$$

$$i(\infty) = -2A, i(t) = -\frac{v(t)}{5} = -2(1 + e^{-1.5t})A$$

$$\text{即 } i(t) = \begin{cases} 0 & t < 0 \\ -2(1 + e^{-1.5t})A, & t > 0 \end{cases} \quad v(t) = \begin{cases} 20V & t < 0 \\ 10(1 + e^{-1.5t})V, & t > 0 \end{cases}$$

例7-12

$$\text{当 } t < 0 \text{ 时, } i(0^-) = \frac{10}{2} = 5A, i(0) = i(0^+) = i(0^-) = 5A,$$

$$\text{当 } t > 0 \text{ 时, } i(\infty) = \frac{10}{2+3} = 2A, R_{Th} = 2+3 = 5\Omega, \tau = \frac{L}{R_{Th}} = \frac{1}{5} = \frac{1}{15}s$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}} = 2 + (5-2)e^{-15t} = 2 + 3e^{-15t}A, t > 0$$

练7-12

$$\text{可得 } I_0 = 6A, i(\infty) = 4A, R_{eq} = 5 + 10 = 15\Omega, \tau = \frac{L}{R_{eq}} = \frac{1.5}{15} = 0.1$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-(t-t_0)/\tau} = 4 + 2e^{-10t}A, t > 0$$